



National University of Computer and Emerging Sciences

FYP-1 Midterm Report

“HireUp” (Development)



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Chapter 1- Introduction

Problem Statement

The previous year proved bleak for various industries and companies in Pakistan, with several expecting to shut down and suffer huge losses this year [1]. Amidst this crisis, arises the need to cut down on various expenses, including the cost spent during hiring processes in the business sector. The average price of the hiring process has risen to **\$4,000** [2]. There are several methods involved in a hiring process which take a lot of time and cost to be executed. Unfortunately, more than half of the resources spent are wasted.

A company typically receives thousands of CVs once a job opening is announced. The HR department has to review each resume and filter the most qualified ones among all applicants. Out of the thousands, at least a hundred applicants are selected for interviews. The interviewing process can take days to conclude. Sometimes, some applicants are called for multiple interviews to ensure their credibility. However, the problem lies in the consumption of time and resources during this process. Electricity costs are multiplied and a portion of the company's assets are consumed. Most of all, time is money. These assets and time can be utilized for some productive activities within the company.

Lastly, a common concern regarding the hiring process is nepotism and biases. In many situations, competent and eligible applicants for a job vacancy fail to secure an interview. Discriminatory practices in recruitment and hiring are an ongoing issue that is a concern not just for workplace relations, but also for a wider understanding of economic justice and inequality [3]. This can lead to unfair selections, completely neglecting the merit.

Project Scope

Users search for the HireUp Website Application on the browser. They will open the application and be able to log in as a recruiter or an applicant. They can sign up to get access to the application if they have not already created an account.

As applicants, users will be asked to enter personal information, including email, phone number, date of birth, and location. Once they have entered these fields, they will receive a confirmation email. After confirming their personal information, they will enter their education, skills, years of experience and desired level of job into the system. This completes the sign-up procedure.

Applicants will receive job suggestions according to their domain and location which they can filter as needed. Each job suggestion will come with an option to apply. A complete description of the job will be shown upon clicking the apply button. This page will guide the applicants to another page where they will fill in their information in a form and upload their cover letter. Applicants can share their LinkedIn URL

and additional information about their experience in the field such as the tools and technologies they are familiar with. The LinkedIn URL will be used to verify their experience.

If the information provided by the applicant complies well with the job description then an interview will be scheduled for them by HireUp. The company they are applying to will conduct automated video interviews or knowledge-based MCQ tests. Finally, applicants will be shortlisted based on their scores in their interview or test.

Applicants will receive notifications about job openings, scheduled interviews/tests and the status of their application. Another feature of HireUp will allow applicants to track the progress of their application at every step.

Recruiters can sign up on HireUp by providing their company's name, email address, contact information and desired number of employees. They will be able to post job openings in their company on HireUp. HireUp will market these openings to potential job applicants. The recruiter also has to choose the final assessment option they prefer for the job opening: an automated video interview or an MCQ-based test. Once a job is posted, recruiters will be notified about the number of applications received and those filtered. They will also be able to manually access the CV, test score or interview recordings of shortlisted applicants. A complete history of their job postings will also be accessible to them.

Modules

1. CV Filtration

This module helps recruiters receive a list of candidates whose CV matches the most with the description of the job opening they have posted.

2. Video Interview Analysis

This module analyzes the video interviews of the shortlisted applicants in order to further assess their eligibility for the job. The accuracy of their speech according to the asked questions and their facial expressions are scored.

3. Knowledge-based Test

This module enables the shortlisted candidates to be further assessed via a knowledge-based test which consists of MCQs.

4. Job Posting & Management

This module enables recruiters/companies to post job openings on HireUp and use this automated tool to find shortlisted applicants for the vacancy.

5. Job Application Submission

This module enables applicants to apply for jobs through HireUp by uploading their CVs. This initializes the application process when applicants search for a job or receive a job suggestion tailored to their interests.

6. Job Search & Recommendations

This module allows applicants to search for jobs tailored to their desired locations and domains. The system also shows job suggestions to applicants tailored to their domains and locations.

7. Interview & Test Scheduling

Through this module, the system will schedule a test or an interview for the applicant if their CV has been filtered in the initial filtration stage. The recruiter has to choose if shortlisted applicants are further assessed through a test or an automated video interview.

8. Application Status Notifications

This module notifies applicants about the progress of their application. It notifies them if the application has been accepted or rejected and also notifies them about their test/interview schedules.

9. Application Tracking

This module enables the applicant to track their application progress.

10. Profile Management

This module enables applicants/recruiters to update personal information in their profiles. The system has to accurately save the changes made. The system should also allow applicants/recruiters to delete their profiles anytime.

User Classes and Characteristics

User Class	Description
Applicant	A primary user of HireUp who is seeking a job and can apply for job openings through the application.
Recruiter	A primary user of HireUp who can find most eligible candidates for a job through the application.

Chapter 2- Project Requirements

Use Cases

Manage Profile

High-Level/Brief Use Case

Use Case	Manage Profile
Actors	Applicant, Recruiter
Type	Primary
Description	Applicants and Recruiters can create, update or delete their profiles. If any information needs to be changed after account creation, the system shall allow the changes to be made and saved.

Fully Dressed Use Case

Use Case	Manage Profile
Scope	HireUp
Level	User goal
Primary Actors	Applicant, Recruiter
Stakeholder and Interests	- Recruiters can update information about their company in the edit profile whenever it is required. They can also delete their profile from the HireUp application at any time. - Applicants can update their personal information in the edit profile menu whenever it is required. They can also delete their profile from the HireUp application at any time.
Preconditions	The applicant/recruiter has logged in.
Success Guarantee	- The changes made to the profile are accurately updated and saved in the database. - If the applicant/recruiter has chosen to delete the profile, it is completely deleted from the database.

Main Success Scenario	1. Applicant/recruiter logs into HireUp. 2. Applicant/recruiter navigates to the “Edit Profile” section. 3. The system displays the currently saved information. 4. The applicant/recruiter makes the required changes. 5. The system validates the changes. 6. The applicant/recruiter saves the changes made. 7. The system displays a confirmation message. 8. The applicant/recruiter confirms. 9. The system saves the changes, updates the database and displays a success message.
Extensions	5a. Applicant/recruiter has made an invalid change. • The system displays an error message. • Applicant/recruiter corrects the incorrect inputs. 7a. Applicant/recruiter cancels the confirmation. • No changes are saved in the system. 9a. System fails to save the changes. • An error message is displayed. • No changes are saved in the system. • The applicant/recruiter is prompted to try again later.
Special Requirements	The system shall be able to update and save the changes within 5 seconds.
Technology and Data Variations	None.

Upload CV

High-Level/Brief Use Case

Use Case	Upload CV
Actors	Applicant
Type	Primary
Description	Upon choosing the option to apply for a job opening, the applicant will be guided to a form where all the relevant fields are displayed. The applicant will fill in all the fields on the form and click on the upload CV button to complete the application process.

Fully Dressed Use Case

Use Case	Upload CV
Scope	HireUp
Level	Sub function
Primary Actors	Applicant
Stakeholder and Interests	Applicants can apply for a job opening by uploading their CV.
Preconditions	- The applicant has logged in. - The applicant has searched for a job or received a job suggestion. - The applicant has navigated to the apply button on the job posting.
Success Guarantee	- All the information about the applicant including personal, career and experience details is saved in the system. - An applicant has applied for the job. - The system has saved the applicant's CV with the job that has been applied for.
Main Success Scenario	- Applicant logs into the system. - Applicant searches for a job or receives a job suggestion. - Applicant navigates to the apply option. - The system displays a form that contains fields required for a CV. - Applicant fills information in each field. - The system validates the information. - Applicant navigates to the upload CV option. - The system displays a confirmation message. - Applicant confirms the details entered in the form. - The system saves the applicant's CV for the job that has been applied for.
Extensions	3a. Applicant does not click on the apply button. - The application process does not start. - The CV form is not displayed. 4a. The system fails to display the CV form due to server error. - The system displays an error message. - The applicant is prompted to try again. 6a. Applicant fills invalid information. - The system shall highlight where the invalid information is filled. - Applicant makes the correction.

	7a. Applicant does not click the upload CV button and navigates to another page. • CV is not saved. • The applicant has to fill out the form again. 9a. Applicant does not confirm the details entered in the form. • The CV is not saved. 10a. The system fails to save the CV due to a server error. • The system displays an error message. • The applicant is prompted to try again.
Special Requirements	• The system should prompt the applicant to fill valid information in the required fields. • The system shall save the uploaded CV within 5 seconds.
Technology and Data Variations	None.

Search for Job

High-Level/Brief Use Case

Use Case	Search for Job
Actors	Applicant
Type	Primary
Description	Applicants can search for a job based on their location or domain.

Fully Dressed Use Case

Use Case	Search for Job
Scope	HireUp
Level	User goal
Primary Actors	Applicant
Stakeholder and Interests	• Applicants can search for jobs according to their skills, location and domain.

Preconditions	The applicant has logged in.
Success Guarantee	The applicant is able to find a job according to the desired location or domain.
Main Success Scenario	- The applicant logs into the HireUp application. - The applicant navigates to the search option on the navigation bar. - The system displays a search portal. - The applicant searches for a job by entering the domain or location. - The system displays job vacancies tailored to the searched requirement.
Extensions	5a. The system does not display jobs tailored to the domain or location of the applicant due to the absence of such job openings. - The applicant is prompted to search again using more terminologies for the desired job. 5b. System does not respond due to server error. - System displays an error message. - Applicants are prompted to try again later.
Special Requirements	- The system should display jobs specifically tailored to the searched domain or location of the applicant. - The system should display jobs within 5 seconds.
Technology and Data Variations	None.

Apply for Job

High-Level/Brief Use Case

Use Case	Apply for Job
Actors	Applicant
Type	Primary
Description	Applicants can apply for jobs that they have searched for or those that have been suggested to them by HireUp.

Fully Dressed Use Case

Use Case	Apply for Job
-----------------	---------------

Scope	HireUp
Level	User goal
Primary Actors	Applicant
Stakeholder and Interests	- Applicants can apply for a job opening. - Recruiters can find a list of the most eligible applicants who applied for the jobs they posted.
Preconditions	- The applicant has logged in. - The applicant has searched for a job or received a job suggestion.
Success Guarantee	- The applicant clicks on the apply button on the displayed jobs according to the desired domain or location. - The system guides the applicant to a CV form, which when accurately filled will initiate the application process.
Main Success Scenario	- Applicant logs into the system. - Applicant searches for a job or receives a job suggestion. - Applicant navigates to the apply option. - The system displays a form that contains fields required for a CV.
Extensions	3a. Applicant does not click on the apply button. - The application process does not start. - The CV form is not displayed. 4a. The system fails to display the CV form due to server error. - The system displays an error message. - The applicant is prompted to try again.
Special Requirements	None.
Technology and Data Variations	None.

View Job Suggestions

High-Level/Brief Use Case

Use Case	View Job Suggestions
-----------------	----------------------

Actors	Applicant
Type	Primary
Description	Applicants are able to view job suggestions displayed to them by the system on the basis of their domains and location.

Fully Dressed Use Case

Use Case	View Job Suggestions
Scope	HireUp
Level	User goal
Primary Actors	Applicant
Stakeholder and Interests	Applicants receive job suggestions from the system that match their domain and location so that they are aware of more job openings suitable for them.
Preconditions	• The applicant/recruiter has logged in. • Information related to the applicant's domain and location is stored in the system.
Success Guarantee	• The system displays job suggestions and the applicant is able to view the suggestions specifically tailored to his domain and location. • The applicant applies filters and filtered job suggestions are displayed.
Main Success Scenario	1. Applicant logs into HireUp. 2. Applicant navigates to the 'Job Suggestions' option on the navigation bar. 3. The system displays job openings tailored to the domain and location of the applicant. 4. Applicants applies filters. 5. The system displays filtered job suggestions.
Extensions	3a. The system does not display job openings due to server error. • The applicant is prompted to try again. 4a. Applicant does not apply any filters. • The same jobs that were initially displayed remain on the page.
Special Requirements	The system shall be able to display the suggestions within 5 seconds.
Technology and Data Variations	None.

Functional Requirements

CV Filtration

- The system shall provide a form to the applicant.
- The system shall extract information such as skills, experience, etc. from the provided data.
- The system shall verify the experience through LinkedIn and GitHub.
- The system shall score the CV based on the requirements of the job description provided by the recruiters.

Video Interview Analysis

- The system shall provide an option to assess the shortlisted applicants via an automated video interview.
- The system shall take questions entered by the recruiter for the interview.
- The system shall question the applicants.
- The system shall record the interviews of the applicants.
- The system shall process and analyze the emotions from the facial expressions of the applicant.
- The system shall analyze the answers provided by the applicant.
- The system shall score the applicant based on the facial expressions and accuracy of each answer.

Test Creation

- The system shall provide an option to the recruiters to create an MCQ-based test.
- The system shall score the test answers provided by the applicant based on the key given by the recruiters.
- The system shall show the test scores of the applicants to the recruiters.

Job Posting & Management

- The system shall provide the recruiters with an option to post a job opening.
- The system shall show the recruiters a history of all the job openings they have posted, including the number of applications received for each vacancy.

- The system shall allow the recruiters to view the CVs, recorded interviews or test scores of shortlisted candidates.

Job Application Submission

- The system shall allow applicants to apply for a job vacancy.
- The system shall allow applicants to upload CVs/resumes and other information.

Job Search & Recommendations

- The system shall allow the applicant to search for jobs.
- The system shall display jobs tailored to the applicant's specified field.
- The system shall suggest jobs to applicants based on domain and location.

Interview & Test Scheduling

- The system shall generate a schedule for applicants, listing all their interviews and tests with the respective timings.

Application Status Notifications

- The system shall notify applicants of any updates or changes in the status of their job applications.

Application Tracking

- The system shall allow applicants to track the progress of jobs that they have applied to, including the acceptance or rejection they have achieved at any stage.

Profile Management

- The system shall provide access to recruiters or applicants through sign-up or login.
- The system shall send a verification email after sign-up.
- The system shall allow the applicants/recruiters to edit their profiles and update the already-mentioned information.
- The system shall allow applicants/recruiters to delete their profiles anytime.

Non-functional Requirements

1. Security Requirements

- Only authenticated users can access the main functionalities of the system.
- User data will be encrypted using AES-256 encryption.

2. Usability Requirements

- Clear and informative error messages are provided to guide users.
- The UI will be innovative and user-friendly, following the User-Centered Design principle to ensure ease of use.

3. Performance Requirements

- The application should respond to user actions within 2 seconds under normal load conditions.

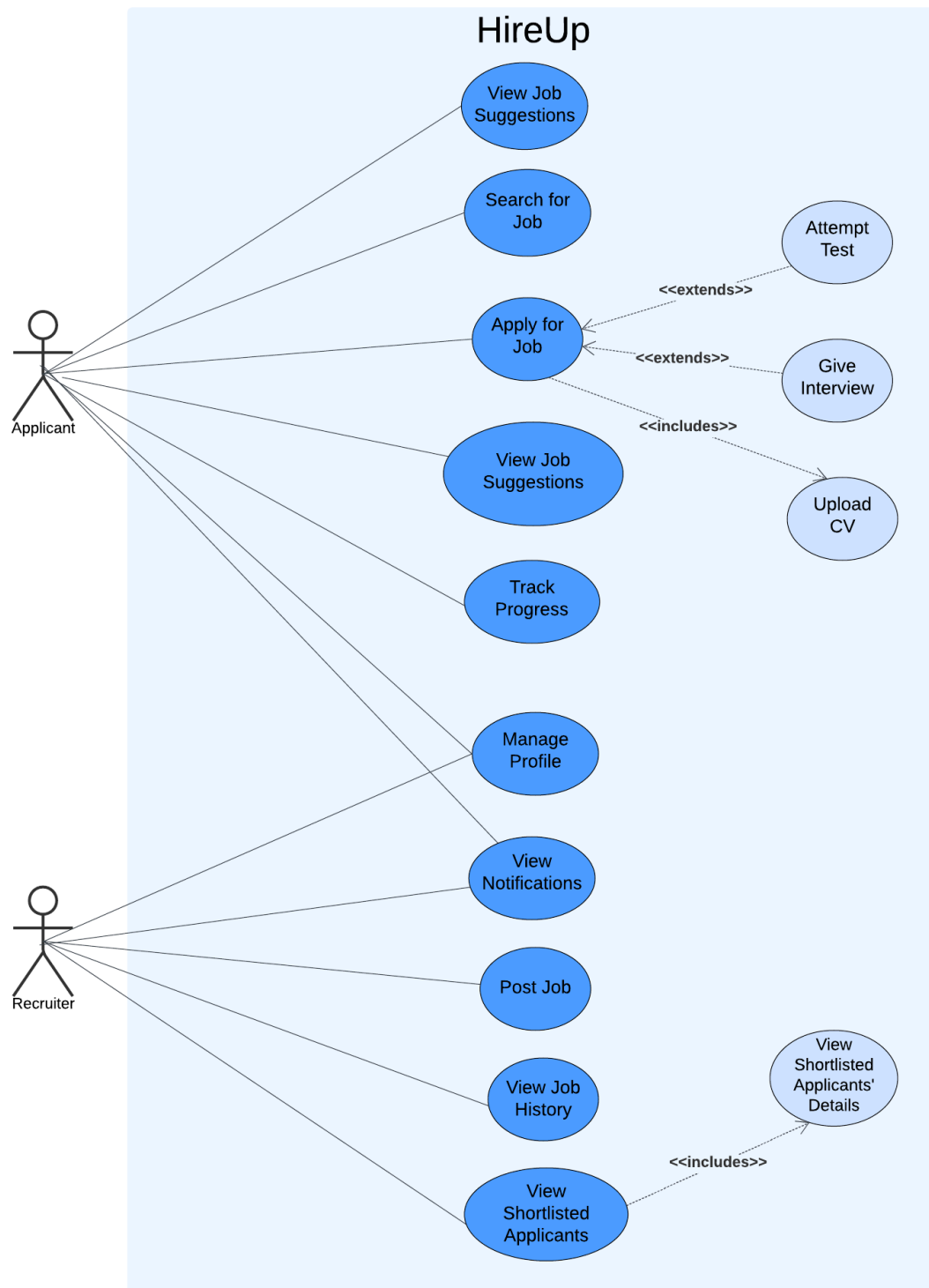
Chapter 3- System Overview

HireUp is a web application that aims to assist recruiters in the recruitment process while also streamlining the job application process for the applicants.

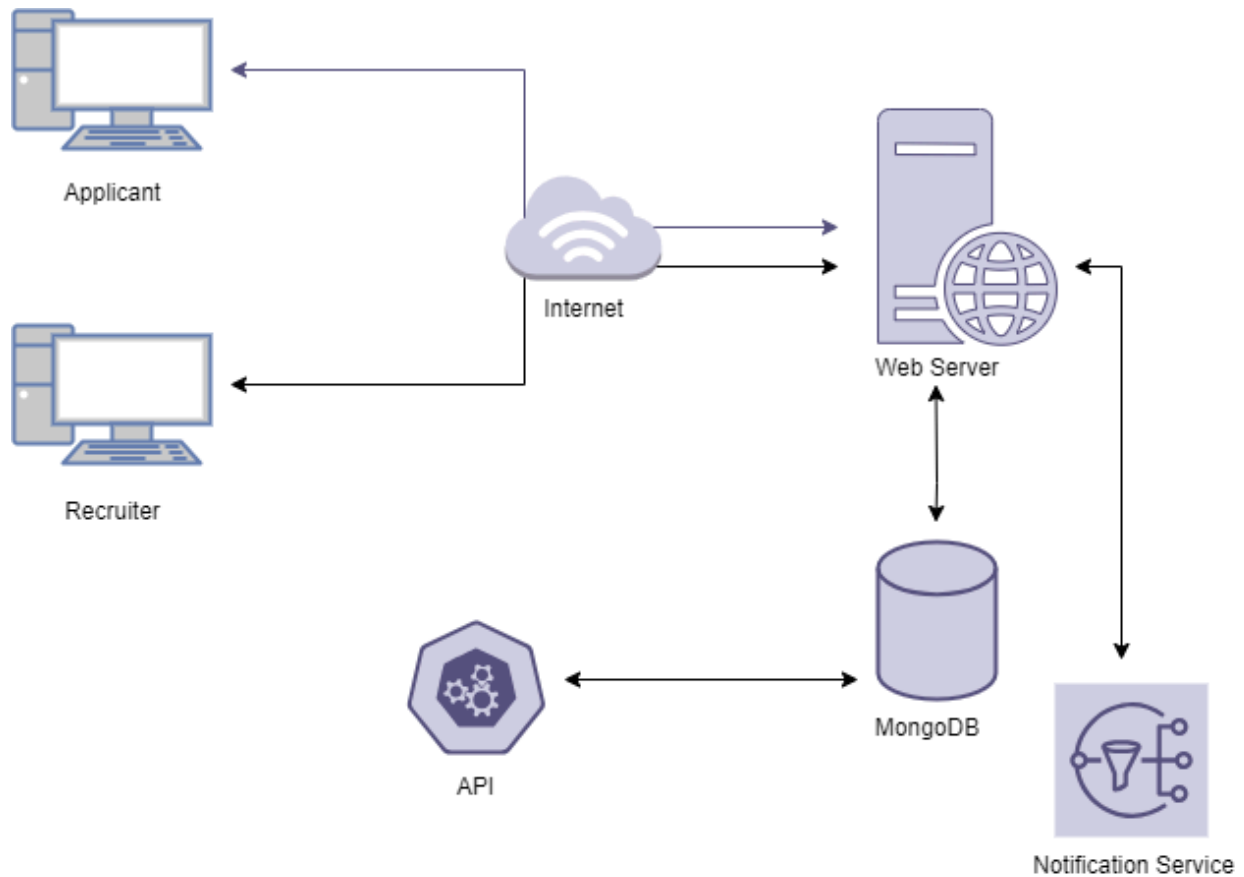
Our key features for iteration 1, are as follows:

1. **Profile management:** Applicants and recruiters alike will be able to edit their profiles to add any other necessary information about themselves or the company and if needed to delete their profiles.
2. **Search for a job:** Applicants will be able to search for jobs either by looking up job titles, company names, skills and location.
3. **Apply for a job:** Applicants will be able to submit their CVs through forms for the jobs they are interested in.
4. **CV filtration:** The system will score and filter the CVs submitted based on the job description.
5. **Job Suggestions:** The system will suggest applicants jobs that are tailored to their domain and location.

Use Case Diagram

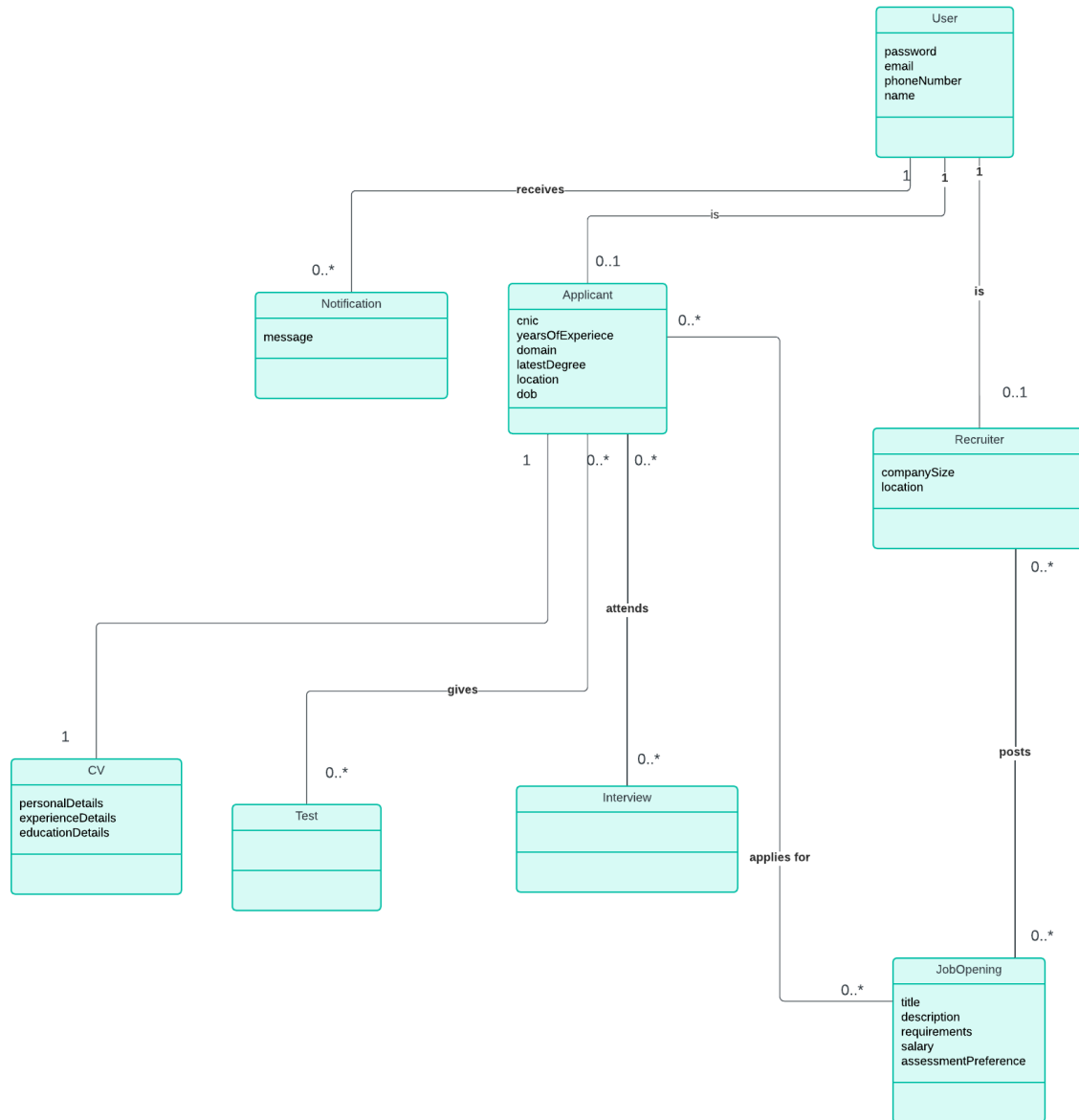


Architectural Design

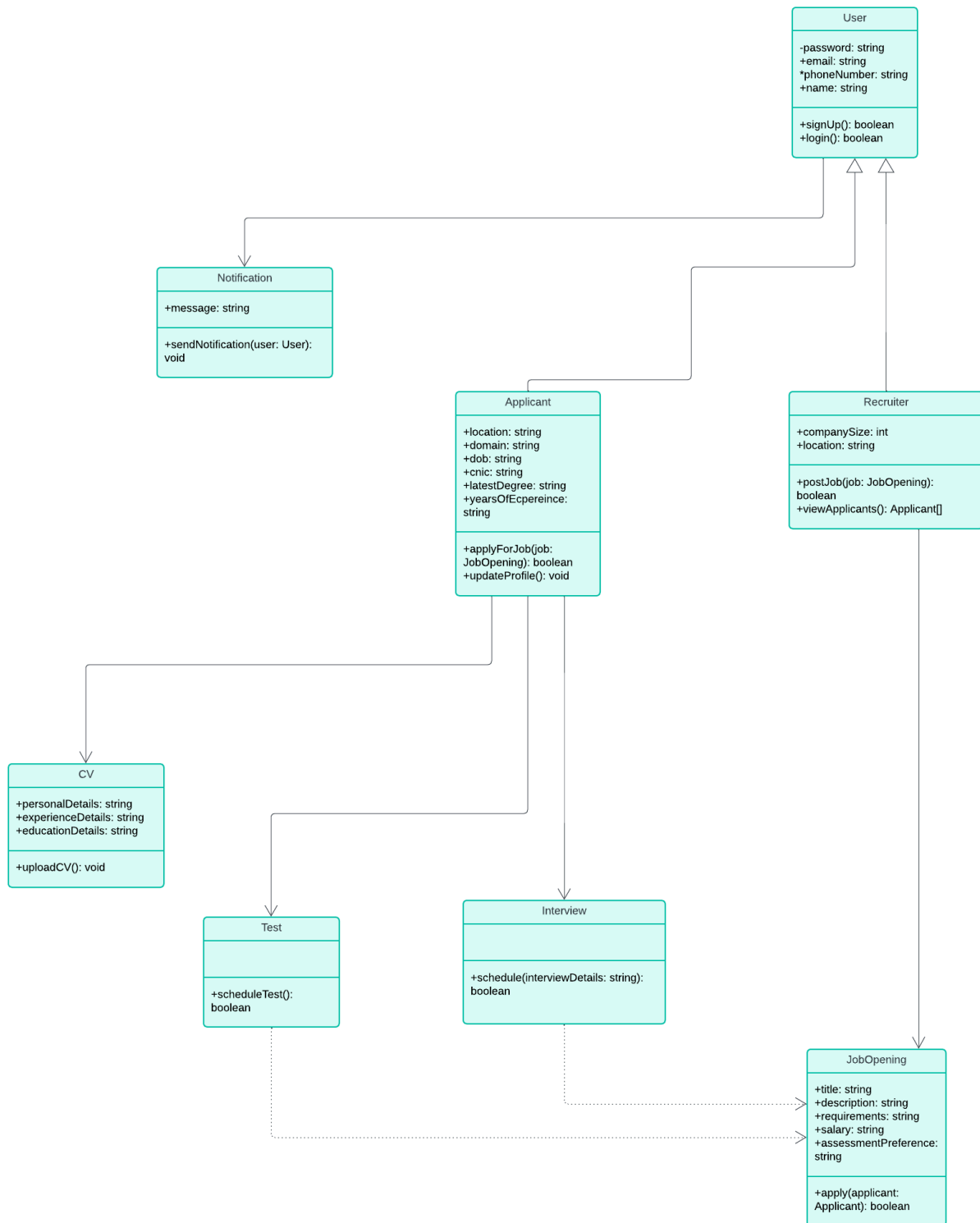


The **client-server architecture** is being used for HireUp. This diagram illustrates the interaction of two kinds of users, applicants and recruiters, with the system by connecting to the internet. The web server handles these requests and uses MongoDB to respond to those requests. The system also utilizes APIs to handle user requests. Notification services alert or update applicants/recruiters about several system actions.

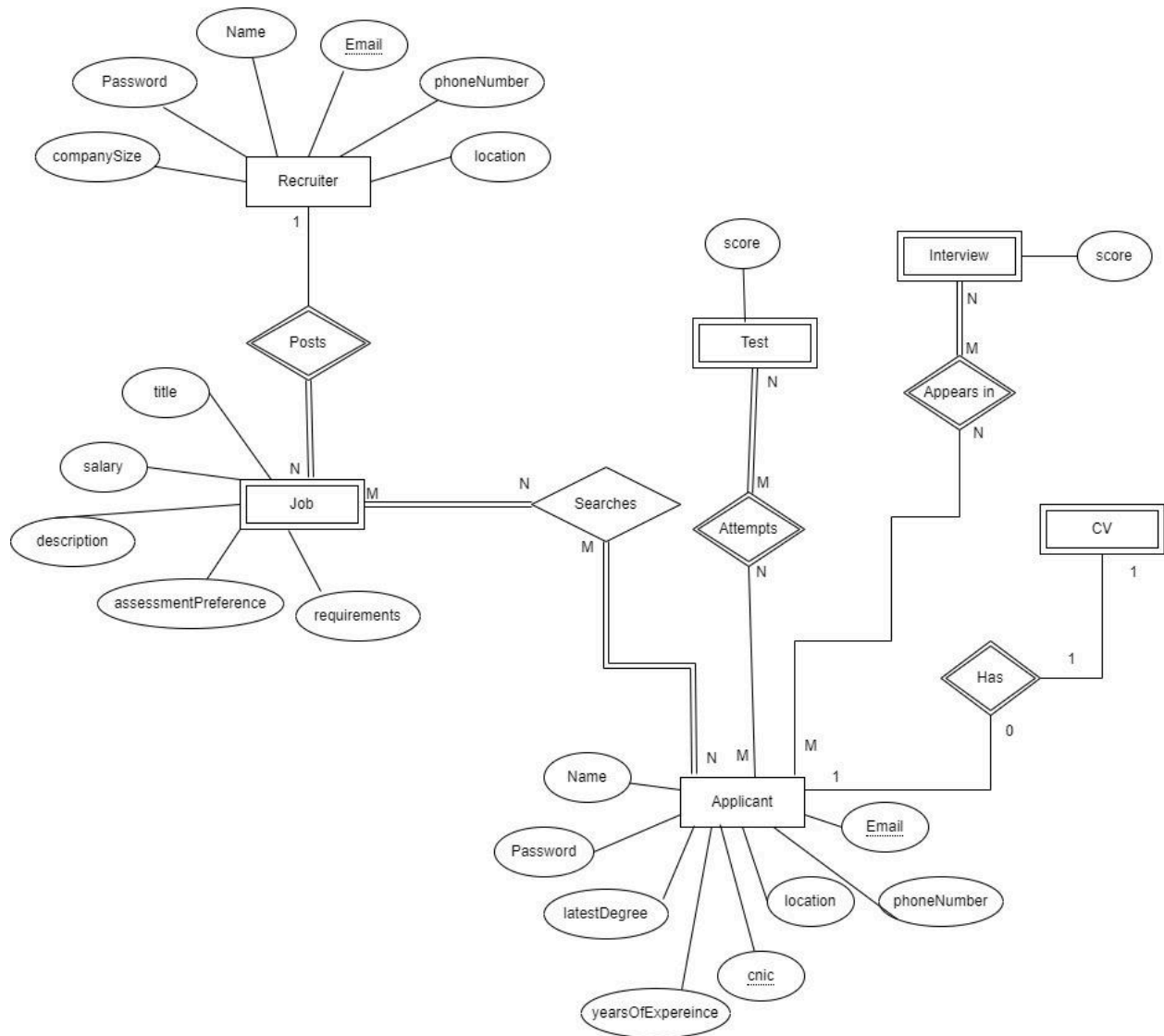
Domain Model Diagram



Class Diagram

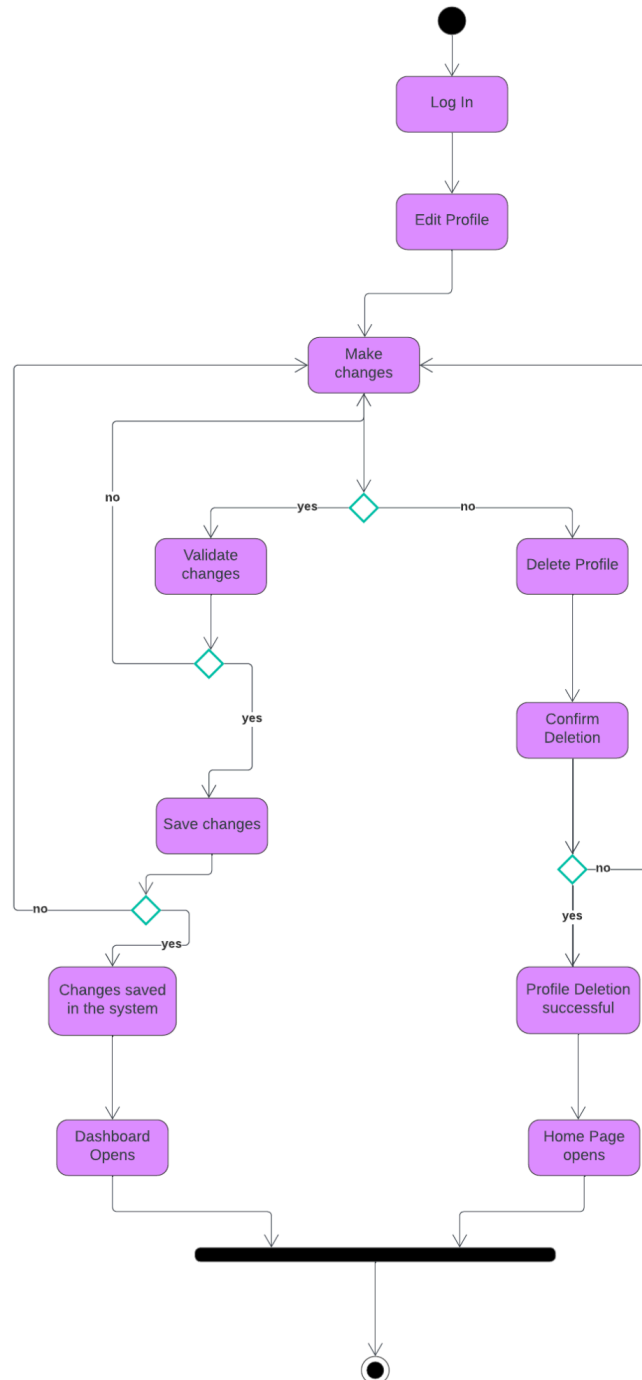


Enhanced Entity Relationship Diagram

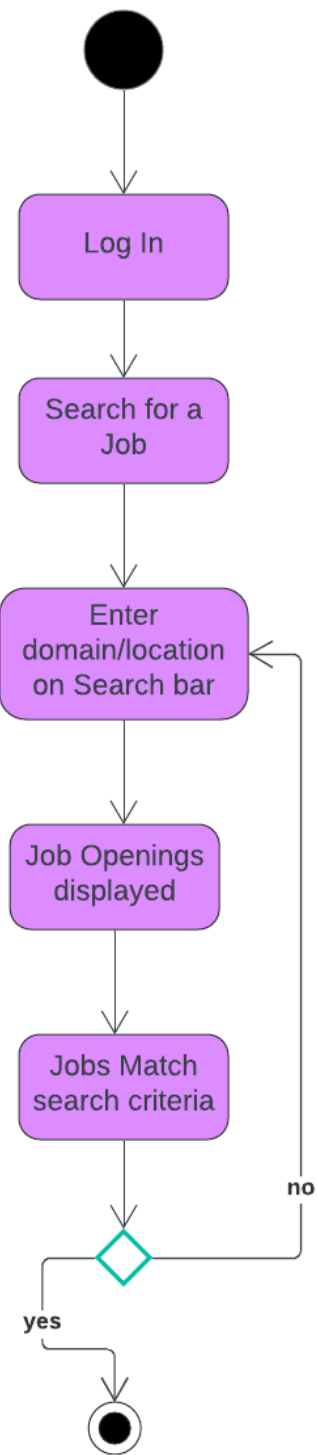


Activity Diagram

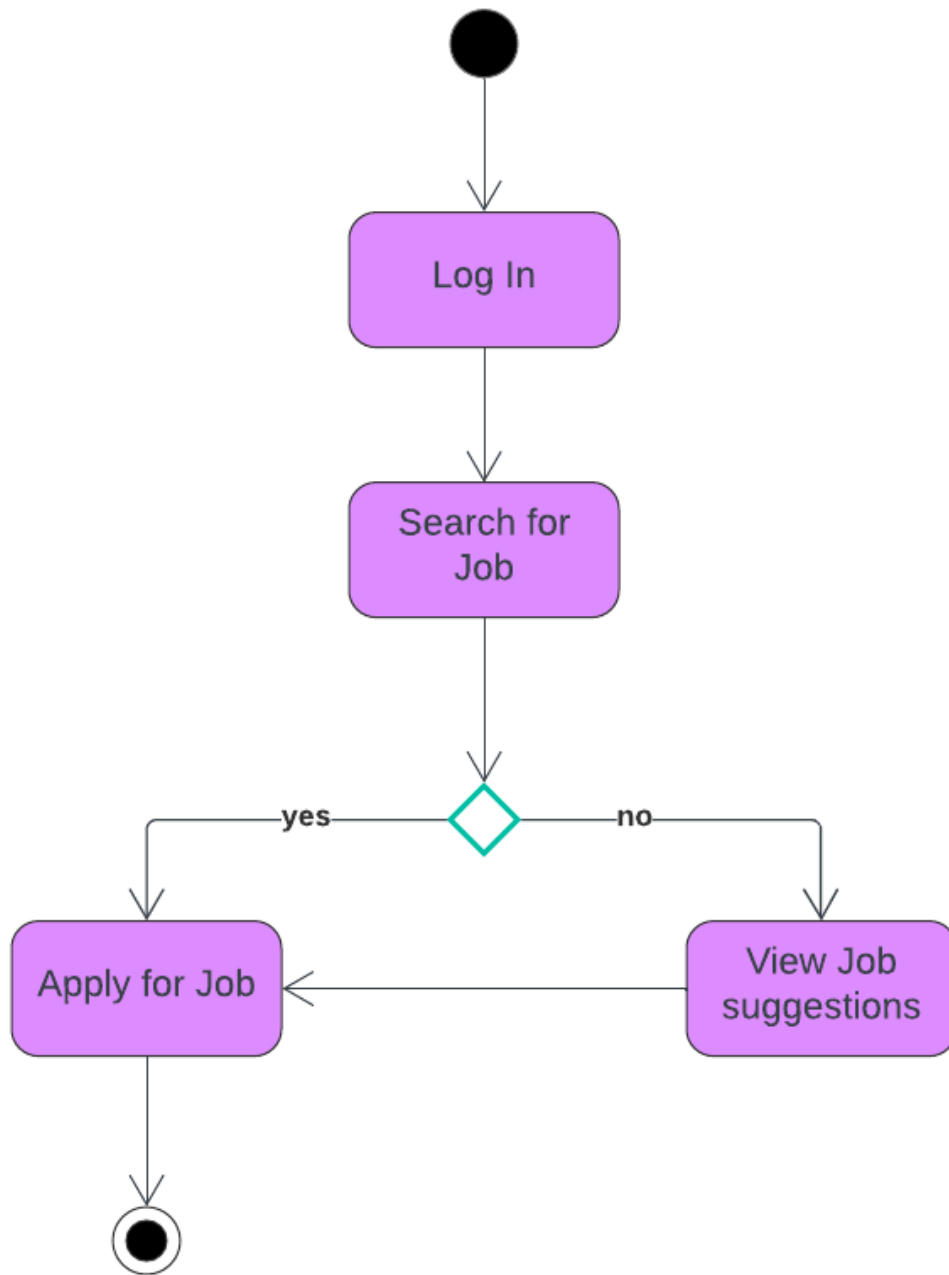
Manage Profile



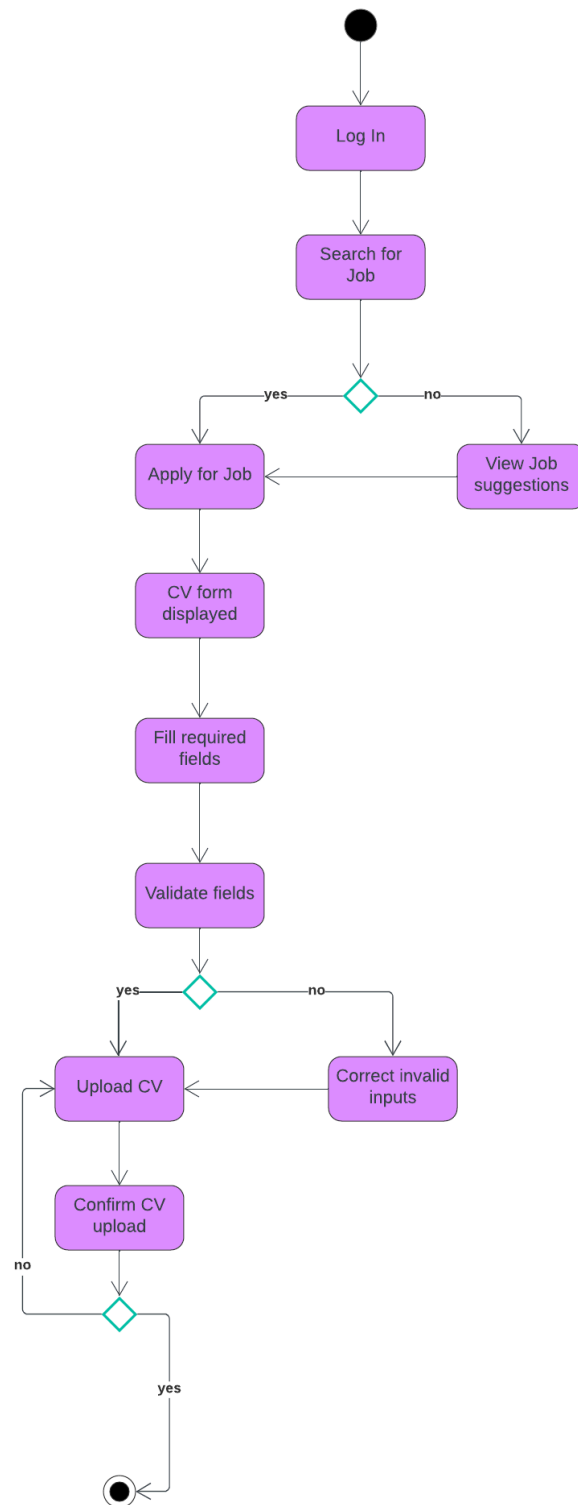
Search for Job



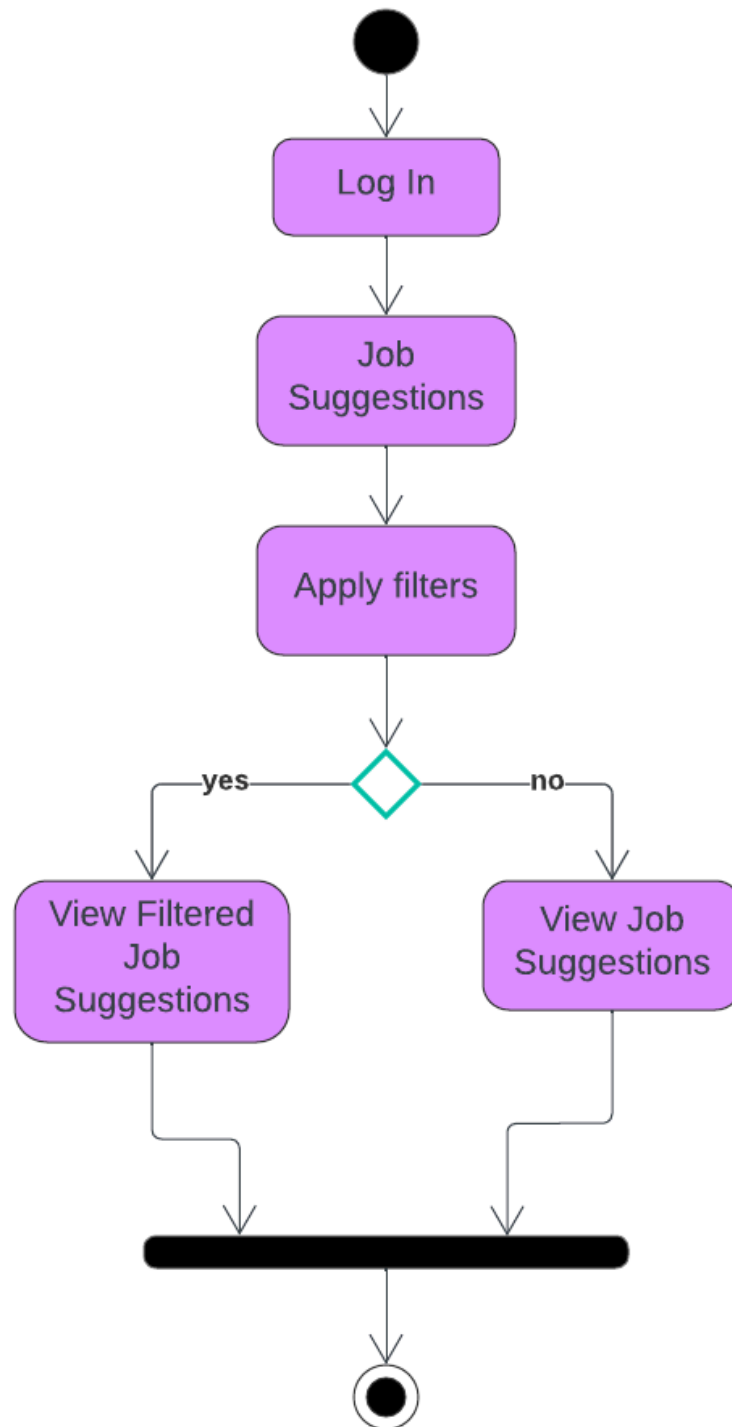
Apply for Job



Upload CV

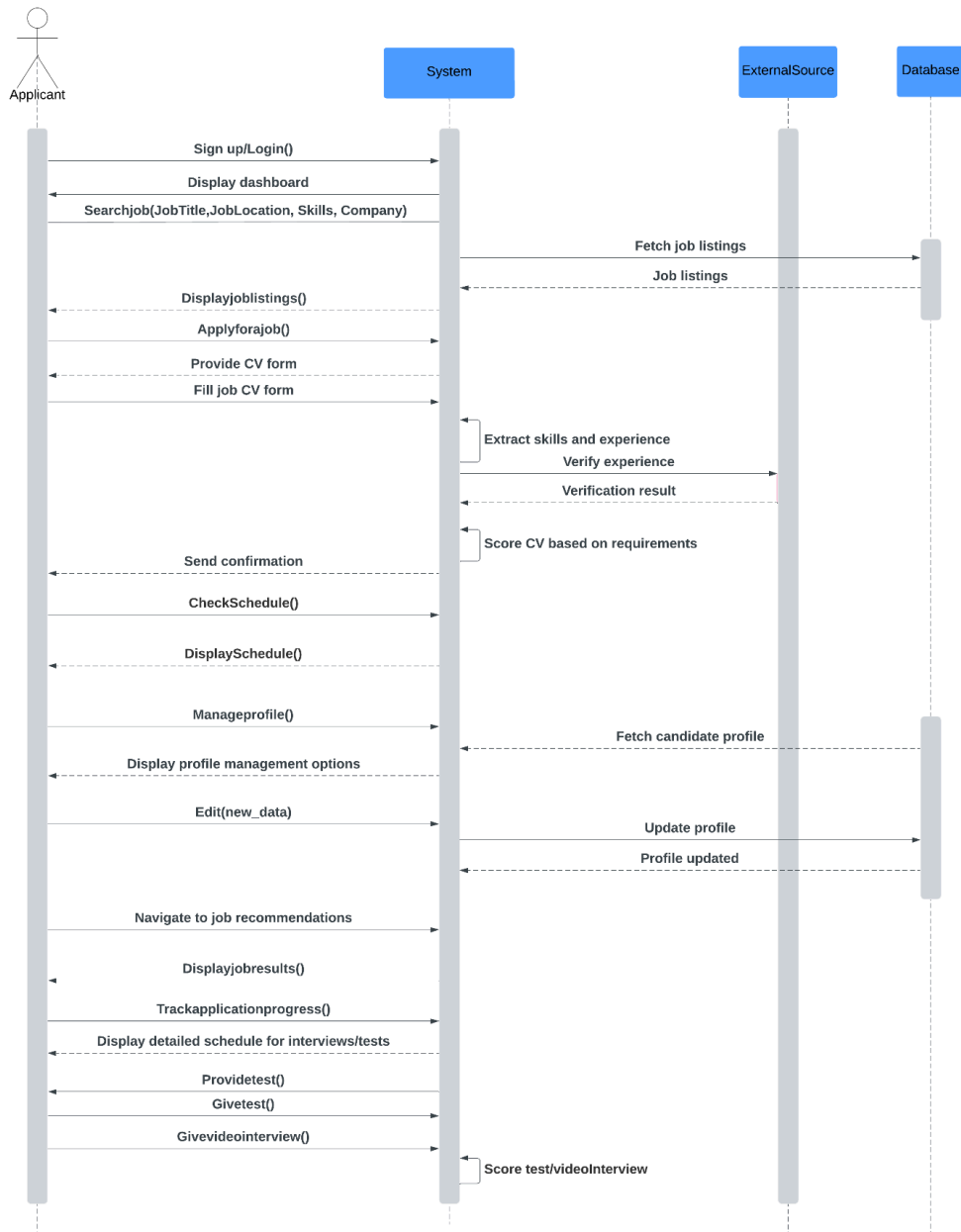


View Job Suggestions

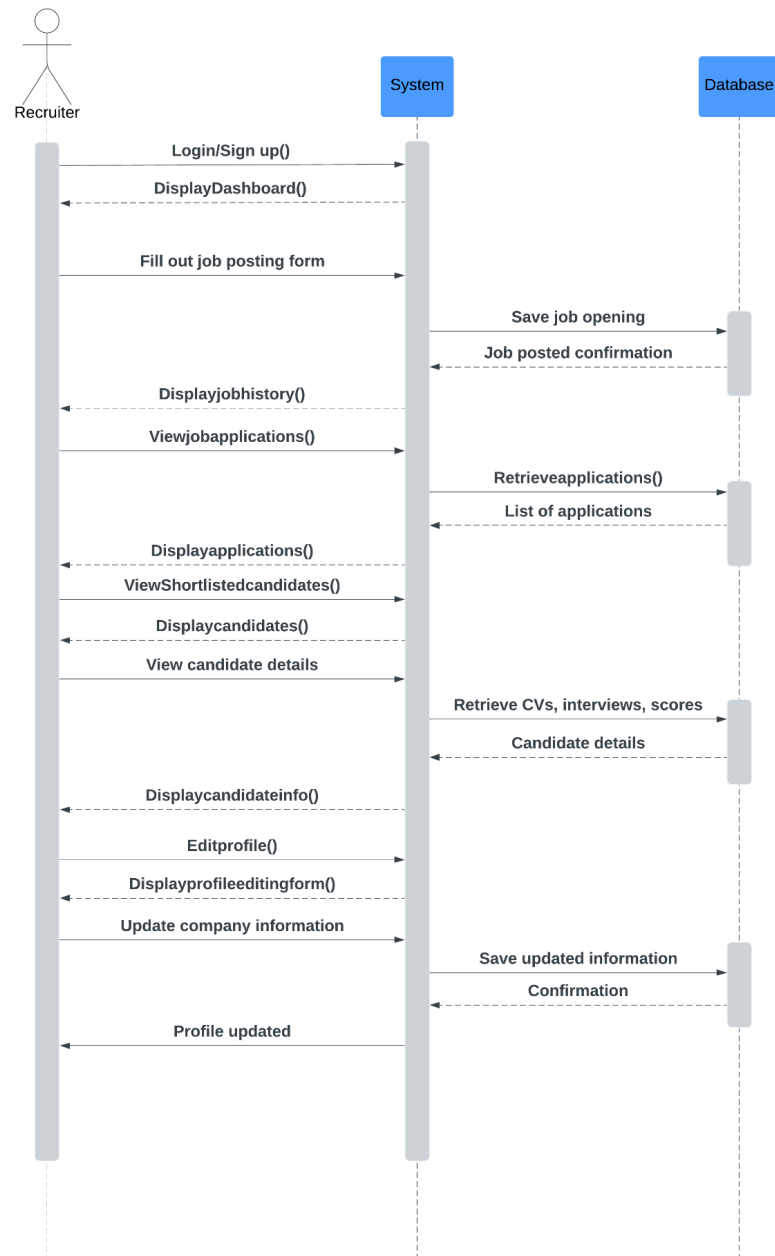


System Sequence Diagram

Applicant View

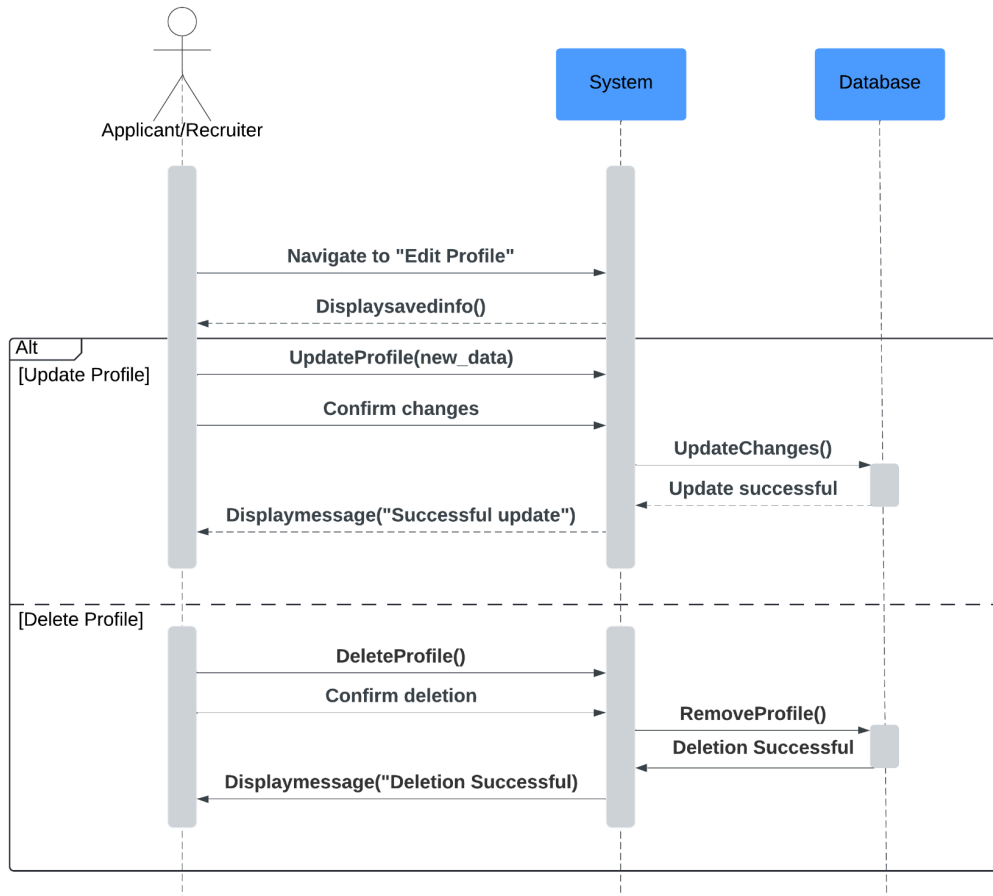


Recruiter View

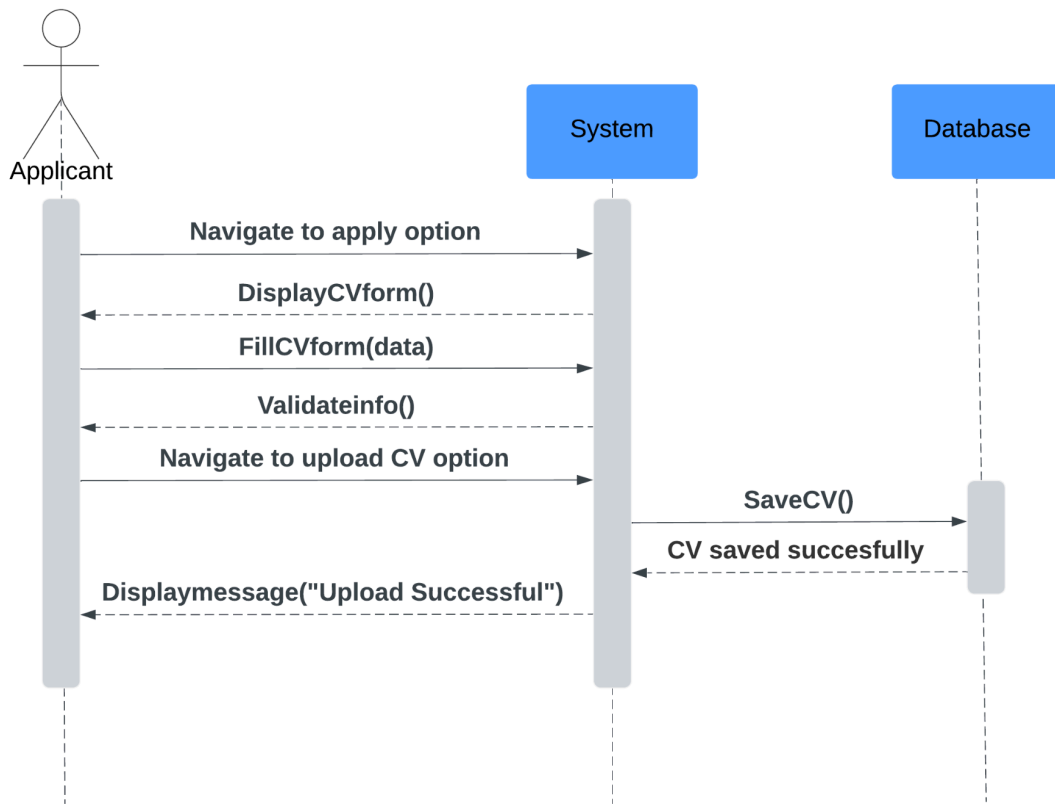


Sequence Diagram

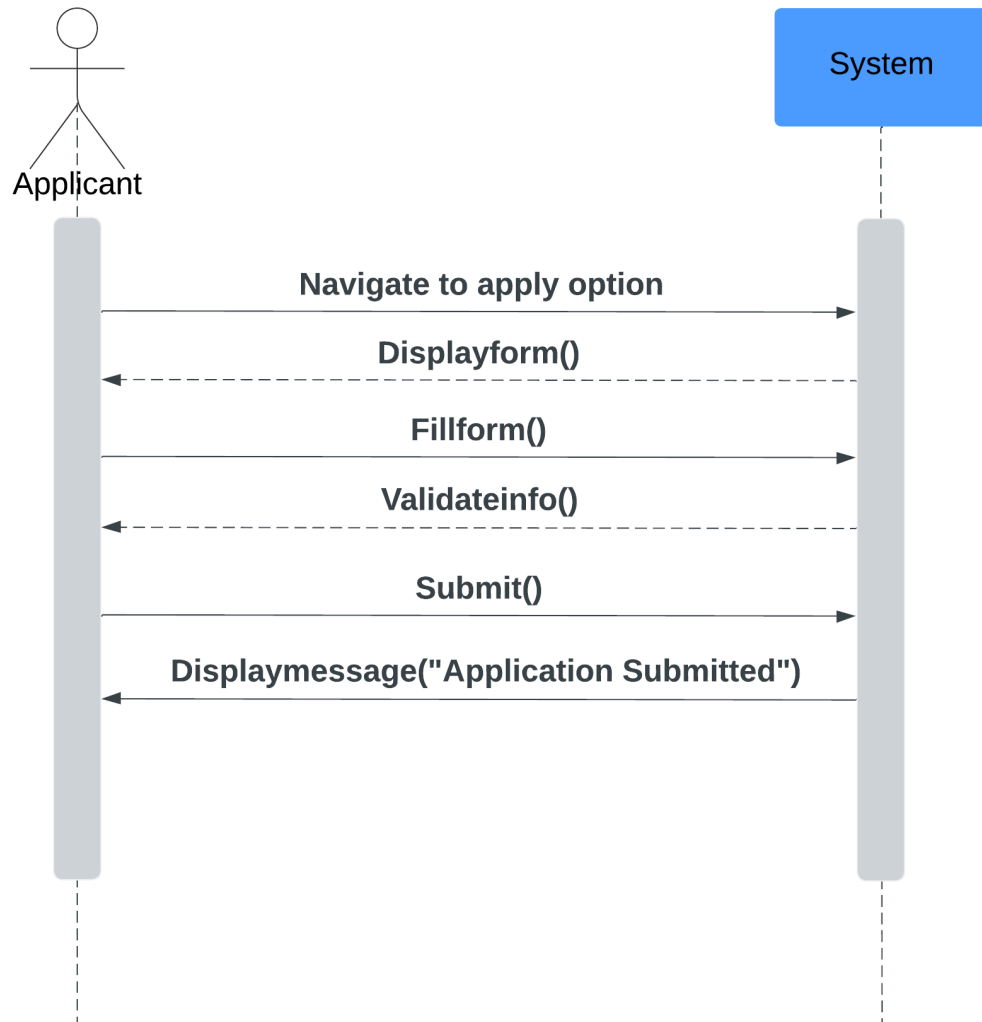
Manage Profile



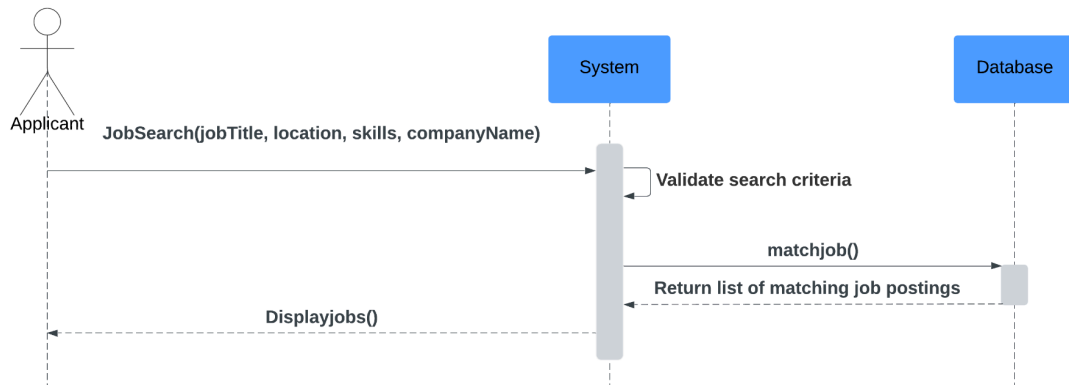
Upload CV



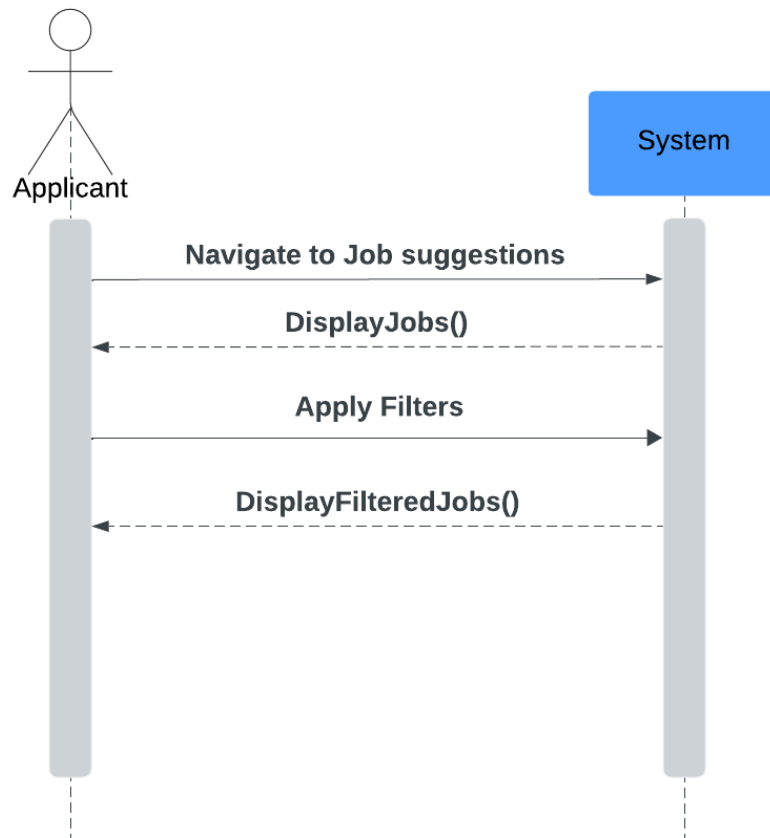
Apply For Job



Search For Job



View Job Suggestions



Prototypes

The following are the prototypes of HireUp:

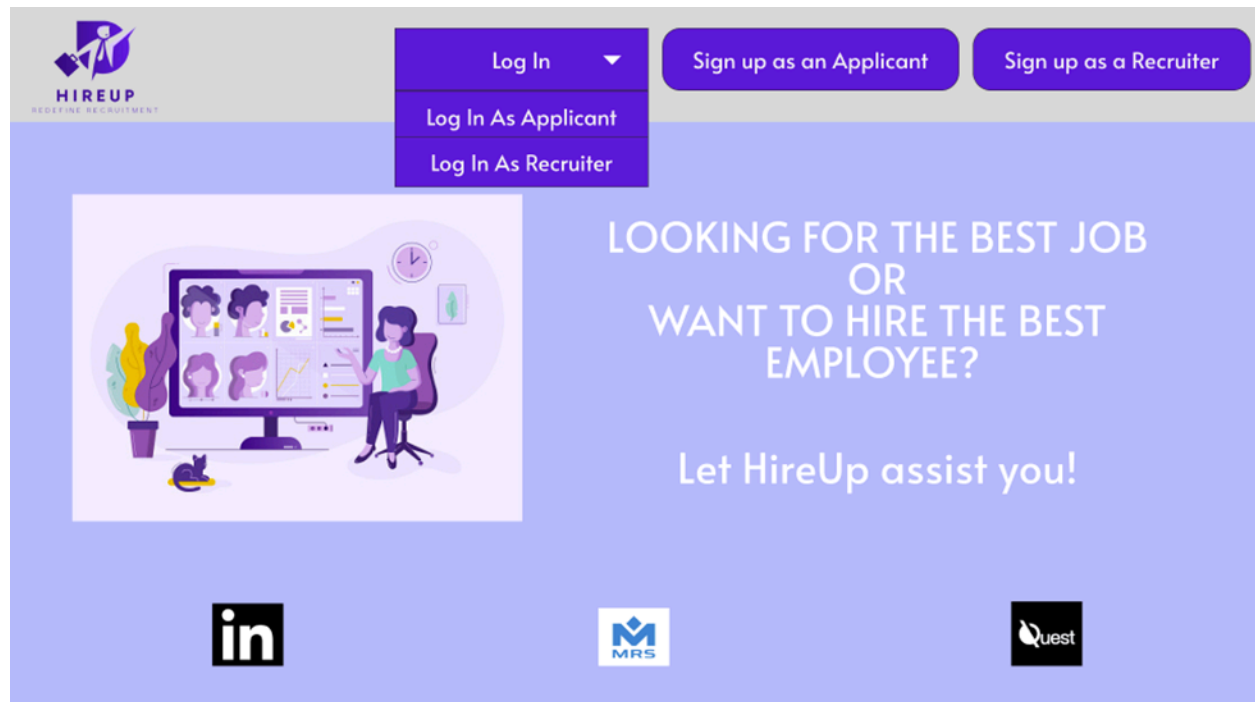


Figure 1: Main Page

Get Started

Full name

Email

Company/Organization name:

Set Password:

City:

Number of Employees:

Best Number to Reach:

Your CNIC (Optional)

[Sign Up](#)

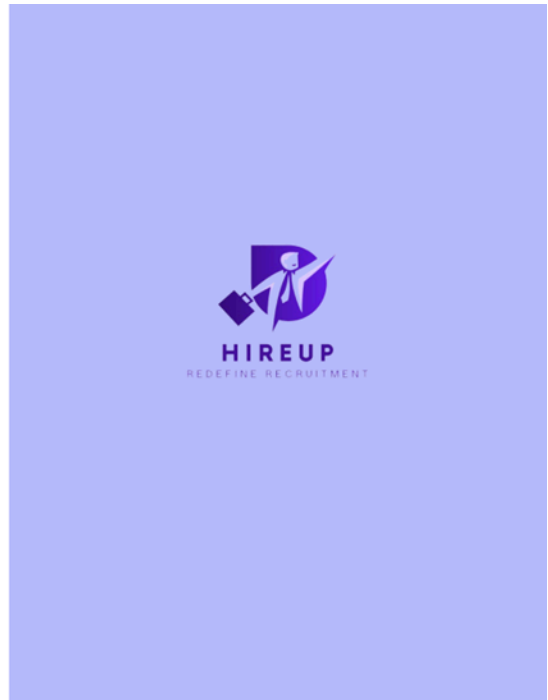


Figure 2: Signup Page for Recruiter/Company

Get Started

Full name

Email

Phone Number

Location

Date of Birth

Latest Degree

Field of Study

Years of Experience

[Save](#)



Figure 3: Signup Page for Applicant

LOGIN

Username

Password

Log In

Create New Account



Figure 4: Login Page for Recruiter & Applicant.

- Home
- Post a Job
- Subscription
- Jobs
- Log Out

MRS Technologies

Objectives:

We build innovative products, digitalize business processes, and speed up development – all using the latest and flexible technologies

We develop scalable products and services that guarantee increased efficiency, improved performance, and greater customer impact

Status

4 Jobs Posted

2 Jobs processed

Notification

Figure 5: Company/ Recruiter Dashboard Page

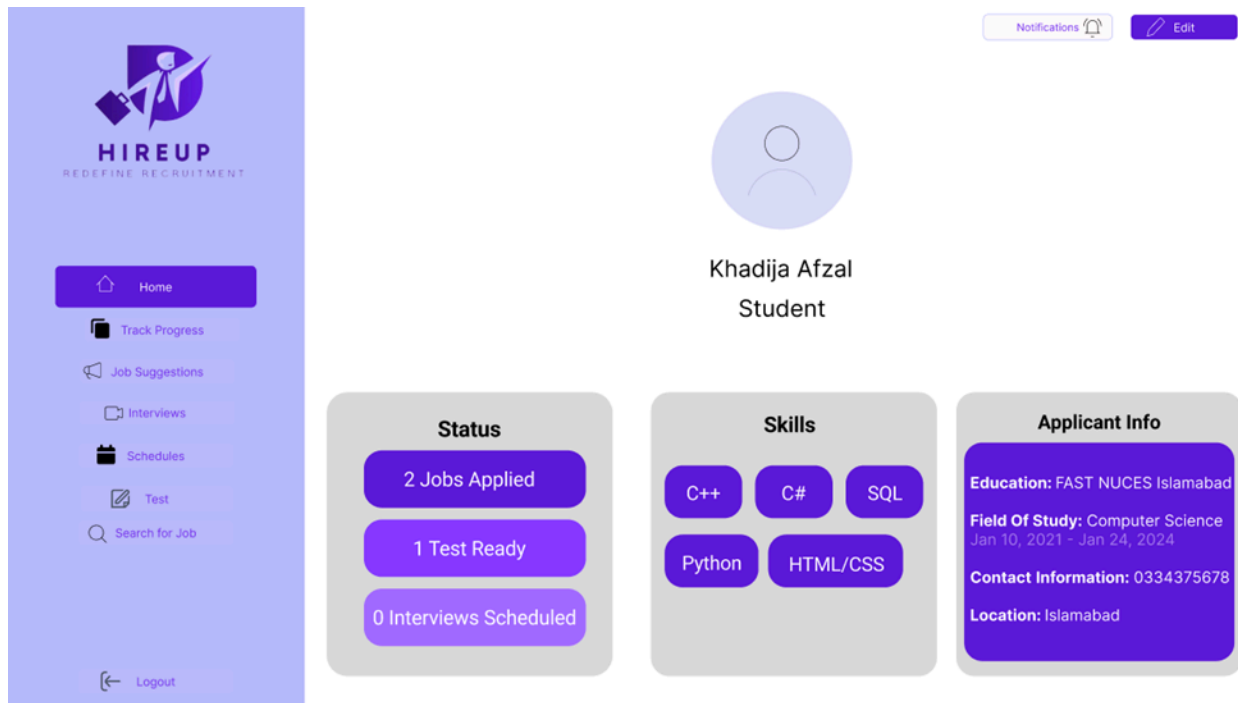


Figure 6: Applicant Dashboard Page

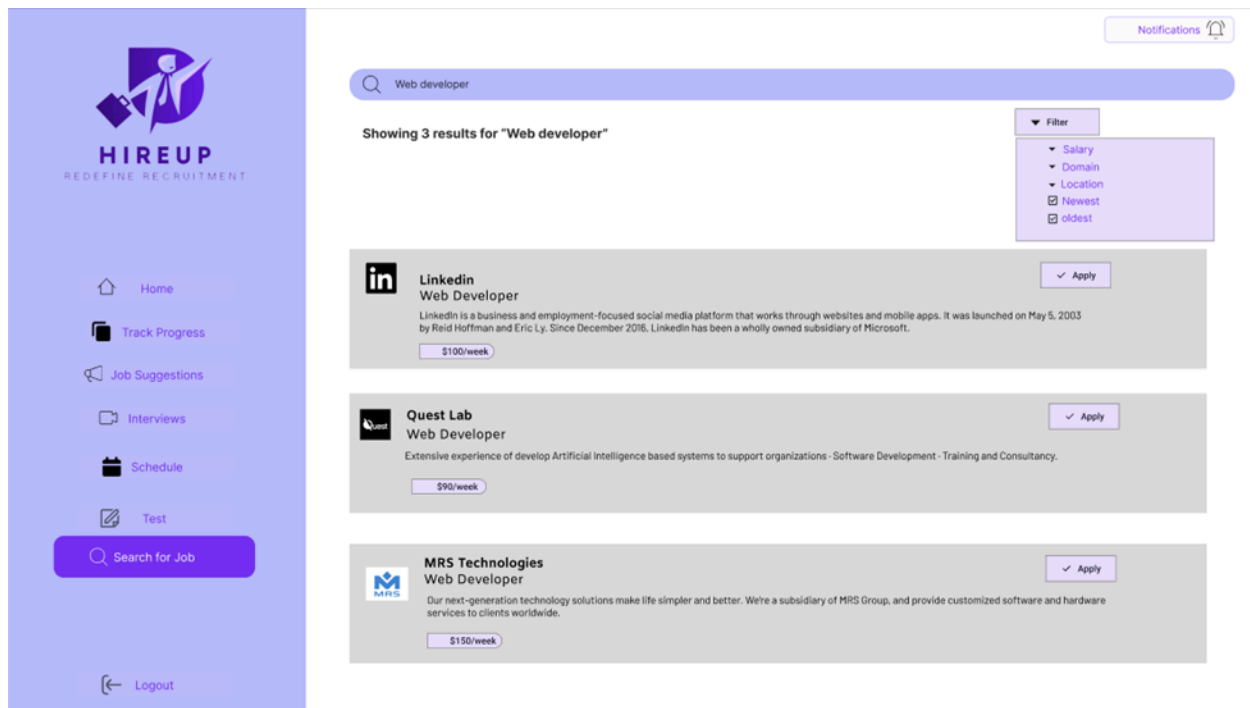


Figure 7: Job Search Page

Fill CV Details

First Name:

Skills:

Add Skill

Experience:

Add Experience

Education:

Add Education

Courses and Certificates:

Add courses/certificates

Interests/Hobbies:

Add Interests/Hobbies

Submit

Figure 8: CV Form Page

Implementation

Profile Management

For this module, we implemented the following functionalities:

- Log in
- Sign up
- Reset password
- Edit profile
- Delete profile

Any changes made to the credentials of applicant/recruiter are updated in the MonogDB database.

Job Search

A set of keywords is stored in the backend which can potentially be used by an applicant while searching for a job. If any of the keywords are found in the job openings' description, those jobs are shown to the applicant in the search results. For example, if an applicant has searched for 'Cyber Security', jobs from this domain will be displayed.

For this module, we have used the following datasets from Kaggle:

Job Suggestions

All keywords that an applicant uses to search for jobs are stored in the database. Jobs with descriptions that include any of those keywords are displayed for the applicant in the 'Job Suggestions' section.

Apply for Job

1. CV Form Page (Initial Form Page)

Objectives:

- Collect essential details such as name, contact information, and education level.
- The user is then navigated to the CV upload section to complete the application.

Core Functionality:

- **State Management:** Utilizes 'useState' to manage input data for form fields and updates data on each input change using 'handleChange'.
- **Navigation:** When the form is submitted, the user is redirected to the '/cv-upload' route for further steps.

2. CV Upload Page (Detailed Form & Submission)

Objectives:

- Capture additional professional details and submit the complete CV.
- Fields such as work experience and expertise in tools are added here.

Core Functionality:

- **Form Submission:** Sends all form data, including the job ID, to the server using a 'POST' request.
- **Confirmation:** Upon successful submission, a confirmation popup appears to inform the user that the application is submitted.

3. CvPopup (Popup Component)

Objective:

- Display a message confirming that the user's application has been submitted successfully.

Core Functionality:

- The popup is controlled by the 'isOpen' state, which decides whether it should be displayed. It disappears when the user closes it.

4. Server.js (Backend)

Objectives:

- Process form submissions from the frontend and store the data in MongoDB.

Core Functionality:

- POST API Endpoint: Receives form data from the client-side and saves it to the MongoDB database.
- Validation & Error Handling: In case the data doesn't meet the schema requirements, validation errors are returned.

Workflow Overview:

1. The user begins by entering basic personal details in `CVFormPage` and clicks "Next" to proceed.
2. On the `CVUploadPage`, the user provides more detailed job-specific information and submits the form.
3. The backend validates and stores the data in MongoDB, and if successful, the frontend shows a confirmation popup to the user.

CV Filtration:

We will be using advanced pre-trained models like BERT (Bidirectional Encoder Representations from Transformers) to rank CVs based on their similarity to job descriptions. Libraries such as HuggingFace's transformers make it easy to utilize these models effectively

Conclusion

In Iteration 1, we aimed to implement the following modules:

- CV filtration
- Job search
- Job Application Submission
- Profile Management

We have achieved our goal by using React, Node.js and MongoDB to implement these modules. We used some pre-existing datasets from Kaggle to perform the functionalities.

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